NOTEBOOK

Ansible - Fundamentals

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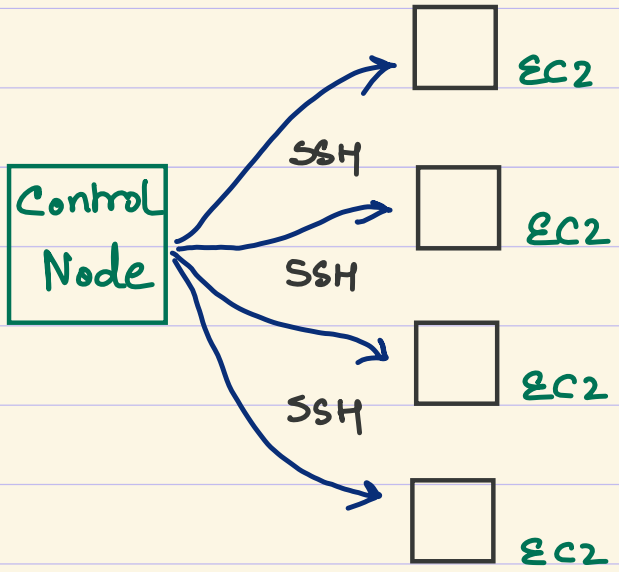
1. What is Ansible

ANSIBLE.

Configuration - management tool. helps to automate setting up infrastructure with essential configuration tuning on multiple resources.



Ansible is agent-less means to configure the receipient we dont to have ansible installed on it.

Supports parallelism :- can run configuration setup simultaneously on multiple resources.

- Save time to market by faster configuration
- Scaletable approach to set-up configurations.

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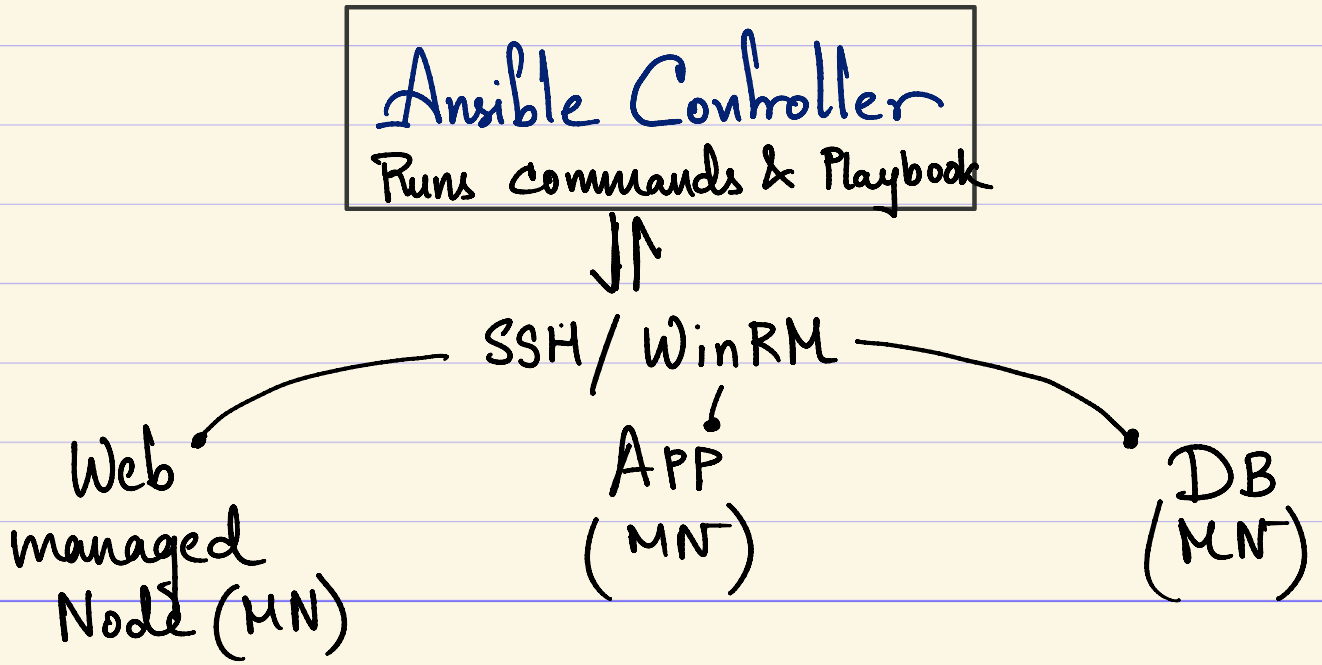
2. Configuration Management:

- Purpose of Configuration management means maintaining Servers in a desired state.

Without Ansible ⟶ Configure Servers manually

With Ansible ⟶ define desired configuration once
↓
Apply to N... Servers.

Ansible Architecture:



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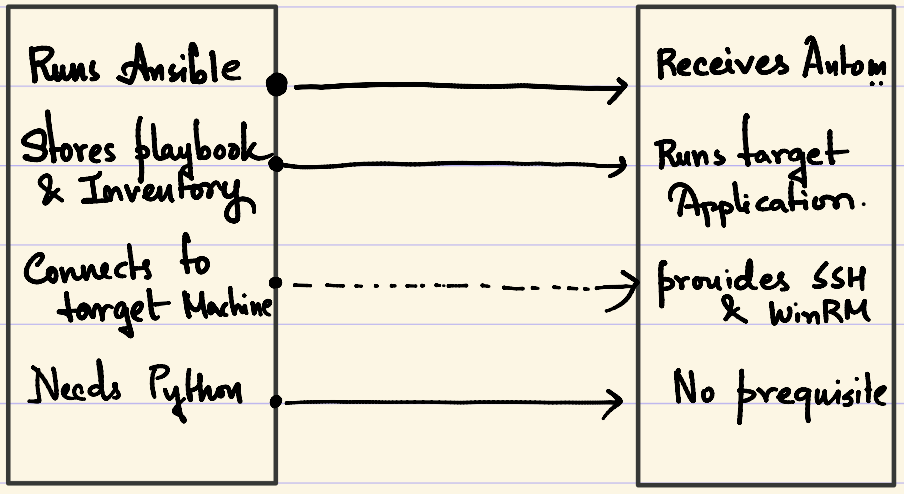
Key Components:

- Controller: Server/Machine where ansible is installed and executed
- Managed Nodes: Server where Ansible configures.
- Inventory: List of target hosts.
- Modules: Performs tasks such as installing packages or Copying files.



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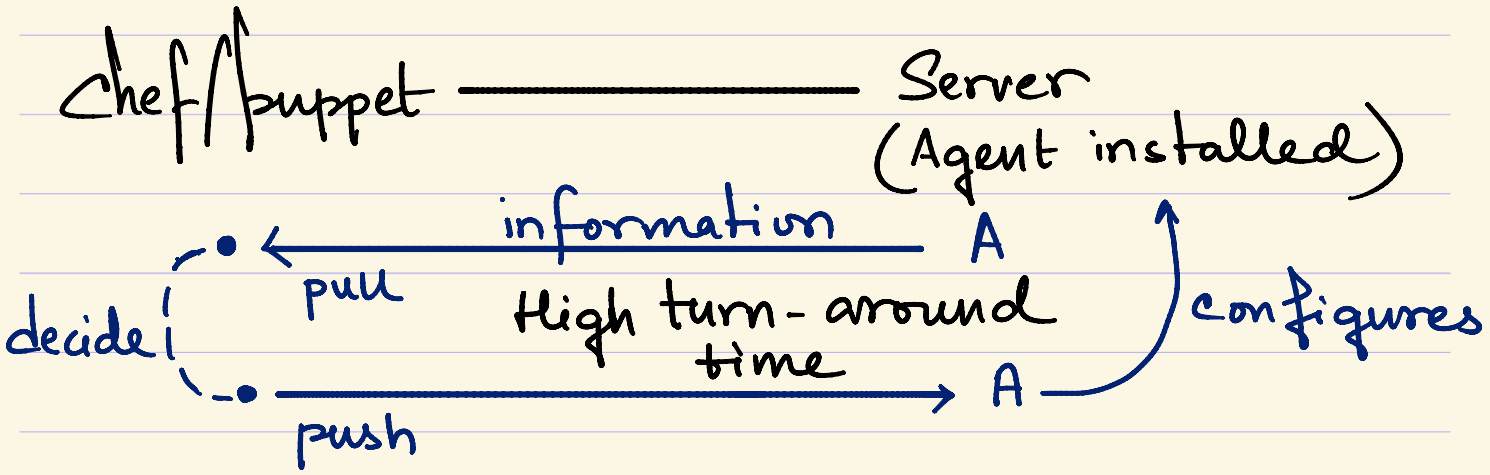
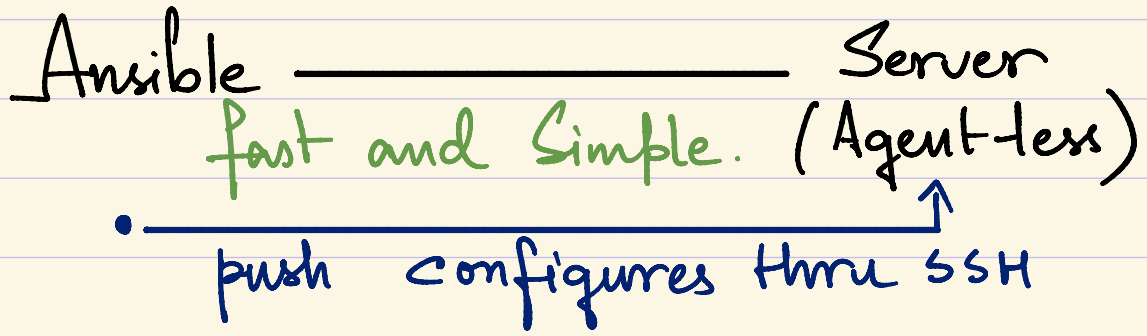
3. How is Ansible different from Chef/Puppet.





4. How to install Ansible

```
Sudo duf install -y ansible
Sudo duf install -y python3
```

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5. Inventory

defines which machines Ansible should manage. Can contain individual IP address, DNS names, group, variables, etc

Example:

group

```
[server]
server-1  ansible_host = <IP-address>
server-1  ansible_user = ec2-user

[all:vars]
ansible_ssh_pvt_key_file = /directory
ansible_python_interpreter =
```

prd

child.

get by which python3

6. Ad-hoc Commands.

Used for executing quick-task without creating playbook

Syntax: ansible <host> -i <inventory file>.ini -m <module> -a "<arguments>" (-v)

verbose

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Commands in details.

-m : modules.
-a : arguments
-b : gives Sudo priviledges. (becomes)

Common Modules.

ping : Test Connectivity

apt, dnf, yum : Managing packages.

Service, systemd_services : Services Management

user, group,

Shell : executes cmd thru shell.

Setup : Gather system facts.

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7. Playbook & tasks:

Playbook is a YAML-file containing plays that define:- which hosts to target and what tasks to perform.

(practical done)

task - <individual operation> inside a playbook, tasks ——calls.——> module

and are generally executed from top → bottom. having a name & action.

8. Variables: Stores values which can be reused inside playbook, inventory, etc.

declaration:

```
vars:
  package_name: nginx
```

↙ to use it

Syntax: "{{ package_name }}"

ansible-playbook -i inventory.ini ↗ override

site.yml | -e "package_name = nginx" |

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9. Facts :

System details Ansible gathers about managed nodes, generally thru setup.

cmd : ansible -i inventory.ini -m setup

10. Registers

Responsible for saving result of a task in a variable ⟶ to inspect later on.

Common Registers.

stdout : command -O
stderr : error
rc : return code
changed / f'd

11. Handlers :

Tasks that only run ↑ when notified by other task

Commonly used to restart/reload Services.

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11. File and Permissions :

file module creates, removes, modifies file or directory

manages : ownership & permissions.

Eg. task

- name : Create directory
  ansible.builtin.file:
    path:
    state : directory / touch / absent → rm
  # makes sure directory / file exists
    owner : ec2-user] ownership
    group : ec2-user]
    mode : 0755 → 0 - R/W/x
    G/oth - R/x.

- Absent / link : remove specified file or dir
  creates symbolic link.

12. Packages & Services :

Packages are software installed on a machine and Services are program running in background.

popular State :

started, stopped, restarted, reloaded [Service config Rt]

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13. Users & Groups:

Ansible can create / modify / delete users and groups → OS.

```
task:
 -name:
  ansible.builtin.group:
   name: devG
   state: present (available)

 -name:
  ansible.builtin.user:
   name: Jenkins11.
   group: devG
   shell: /bin/bash
   create_home: true
   state: present
```

creating user.

```
 -name:
  ansible.builtin.user:
   name:
   state: absent
   remove: true
```

removes a user

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14. Tags:

tags allow you to run selected task or section of playbook.

```
task:
- name: Install Nginx
  ansible.builtin.package:
    name: nginx
    state: present
  tags:
    - install
```

How to run: ansible-playbook -i inventory.ini site.yml --tag install/

--skip-tags → Run all except.

runs only install task

15. Group_vars & host_vars. (directories)

allows us to organise variables outside the playbook, often help in keeping env specific variables outside

[practice]

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16. Roles:

Roles organises Ansible automation into standard directory structure. particularly useful when multiple project need same standardize Server Configuration

[practice]

17. Ansible Galaxy:

Repo - tool for discovering and installing resuable roles and Collection.

↓ listed in requirements.yml

ansible-galaxy collection install -r requirement.yml.

18. Ansible Collections: Standardized packaging format to export / distribute ansible content....

- Components: Packages Modules, plugins, roles, documentation into Single modular structure.
- NameSpace, independent- Versioning, Distribution.

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19. Ansible Vault :

ansible-vault create secrets.yml

Specialized Storage which encrypts Sensitive data : Passwd, API token, Pvt Config....

How to use :

secrets.yml

```
db_pass: * * * * *
db_user : ....
```

playbook :

```
- name : Config db.
  var_files :
    - Secrets.yml

task :
  - name :
    ansible.builtin.debug
      var: db_uses
```

Run :

ansible-playbook -i inventory.ini file.yml --ask-vault-pass

protected Vault : locked with passwd.

--vault-passwod-file.

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20. Error Handling:

Project: Deploy EC2 with terraform and configure with ansible.

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